▶ LUDOVICA CONTI, *Weak Impredicativity.*
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A definition is said to be impredicative if it quantifies over a totality to which the *definiendum* belongs. The debate on impredicativity shows that this term can refer to different phenomena ([4]). In this talk, I will examine various forms of what Gödel termed "weak impredicativities" ([3]) and consider whether they could inform a consistent revision of Frege's logicist project ([5]).

Firstly, I would argue that the notion of predicativity involved in strict and classical predicativism is originally syntactic. Conversely, the notion of generalised or constructive predicativity stems from an ontological characterisation of the structure of predicative entities. This characterisation is very similar to that captured by the set-theoretic criterion of well-foundedness (cf. [2], [1]). I support the idea that the weakly predicative phenomena that Gödel had in mind involve this ontological notion of predicativity, and I propose referring to the others as forms of "strong" impredicativity.

Secondly, I emphasise that the effect of a "weakly predicative" restriction of Frege's system has so far been unexplored, and I aim to demonstrate that it would introduce an interesting modification to *Grundgesetze* because it would prevent the specification of Russell's concept ($Rx \leftrightarrow \exists X(x = \epsilon X \wedge \neg Xx)$), yet remain open to Frege's vocabulary particularly the definitions of ancestral, weak ancestral, predecessor and natural number. Furthermore, it would enable the derivation of second-order Peano axioms, including induction.

[1] Contente M., Klev A., The Nature and Extent of Constructive Predicativity, presentation within CFvW Colloquium, The Carl Friedrich von Weizsäcker Center of the University of Tübingen, November 12, 2025

[2] Crosilla, Laura. "Predicativity and constructive mathematics." In Objects, structures, and logics: FilMat studies in the philosophy of mathematics, pp. 287-309. Cham: Springer International Publishing, 2021.

[3] Gödel, K. (1990). Publications 1938–1974, Volume 2 of Collected Works. Oxford: Oxford University Press. Edited by S. Feferman, J. Dawson Jr., S. Kleene, G. Moore, R. Solovay, and J. van Heijenoort.

[4] Feferman, S. (2005). Predicativity,The Oxford Handbook of Philosophy of Mathematics and Logic

[5] Frege, G. (1893/1903). Grundgesetze der Arithmetik: I.-II. Band (Vol. 1, Vol. 2). H. Pohle.